

Research Article

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Opponent choice in tournaments: winning and shirking

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Abstract: We propose an alternative design for tournaments that use a preliminary stage, followed by several rounds of single elimination play. The conventional “bracket” design of these tournaments suffers from several deficiencies. Specifically, various reasonable performance criteria for the tournament are not satisfied, there is an unnecessary element of luck in the matchups of players, and there are situations where players have an incentive to shirk. To address all these issues, we allow higher ranked players at the single elimination stage to choose their next opponent sequentially at each round. We allow each player’s ranking either to remain static, or to improve by beating a higher ranked player (Guyon, J. 2022. “Choose your opponent”: a new knockout design for hybrid tournaments. *J. Sports Anal.* 8: 9–29). Using data from 2215 men’s professional tennis tournaments from 1991 to 2017, we demonstrate the reasonableness of the results obtained. We also perform sensitivity analysis for the effect of increasing irregularity in the pairwise win probability matrix on three traditional performance measures. Finally, we consider strategic shirking behavior at both the individual and group levels, and show how our opponent choice design can mitigate such behavior. Overall, the opponent choice design provides higher probabilities that the best player wins and also that the two best players meet, and reduces shirking, compared to the conventional bracket design.

Keywords: multiple round sports tournament; choice of opponent; performance criteria; professional tennis data

1 Introduction

We consider multiple round tournaments, as used extensively to organize sports events worldwide. Since our work can be applied to both individual and team tournaments, we use the term “player” to describe either an individual player or a team. Many multiple round tournaments consist of a preliminary or group stage, followed by several rounds of single elimination play using a fixed seeding or *bracket*. This design is used by, for example, most U.S. major sports, the FIFA World Football Cup, and the ICC World Cricket Cup.

The objectives of the designers of such tournaments include: providing the players with equally fair opportunities, motivating them to perform well, selecting among them, and providing an appealing event for spectators. Our work addresses several deficiencies that arise in such tournaments with respect to these objectives, as we now discuss.

First, as documented in the tournament design literature discussed below, various reasonable criteria such as stronger ranked players having a higher probability of winning than a lower ranked player, are not satisfied. Second, the probability that the top two players meet in the final round is not maximized. Third, there is the widely observed issue of *shirking* or *tanking* at the preliminary stage, where a player deliberately avoids winning a game in order to obtain an easier path through the tournament. This occurs where a preliminary stage match can be lost, but the player continues to the next round anyway. Fourth, top ranked players randomly incur unfortunate matchups against other players, which introduces an unnecessary element of luck into the tournament. We believe that the achievement of a top ranking at the preliminary stage should not disadvantage a player based on chance. Finally, the use of a conventional fixed bracket fails to allow players to consider information that develops during the tournament, such as injuries to other players. The conventional fixed bracket design does not allow for this developing information to be used. This is a problem of *dynamic scheduling*, where information changes during the execution of some previously scheduled tasks. Suresh and Chaudhuri (1993) provide a survey of issues and solution approaches for dynamic scheduling. As an example of an important class

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of dynamic scheduling problems, Mohan et al. (2019) discuss dynamic scheduling strategies for job shop problems.

To address these deficiencies of conventional tournament design, we recommend a new design under which the players, in ranked order that is at least initially determined at the preliminary stage, *choose their next opponent at each single elimination round*. As we discuss in Section 2, Guyon (2022) proposes a similar design with a different objective. The choice of opponent is made under the two conditions that the opponent has not been previously chosen by a higher seeded player, and that the player has itself not been previously chosen. This opponent choice design allows considerable flexibility in implementation. For example, we study the performance of two versions: under *static ranking*, the ranking of the players when they entered the single elimination stage of the tournament remains fixed; whereas under *dynamic ranking*, a lower ranked player inherits the ranking of a higher ranked player which it beats. Flexibility is also available regarding the number of player(s) from the top of the ranking, which are allowed to choose their opponent(s), and to exempt higher ranked players from being chosen. By allowing a superset of choices, the opponent choice design provides an easier path through the tournament to a typical player as a result of obtaining a higher ranking.

Our recommendation for opponent choice is based in part on the perspective that the tournament designer wants better players to have a greater chance of winning, but not excessively so, which we refer to as *balance*. Relevant performance measures that enable the tournament designer to achieve this include: (a) whether the opponent choice design provides a reasonably higher tournament win probability to the top ranked player(s), (b) whether it provides a reasonably higher probability of the top two players meeting, and (c) whether it provides a ranking by tournament win probabilities that closely matches the original ranking of the players. Another perspective is that the tournament designer's main aim is financial success. Tournament organizers typically enhance their financial prospects by expanding tournaments by incorporating additional teams instead of changing the opponent structure. A case in point is the expansion of the FIFA World Cup, which evolved from a tournament featuring 16 teams to 24 teams in 1982, subsequently to 32 teams in 1996, and ultimately to 48 teams as of the 2026 tournament. This strategic expansion of participation reflects a trend among organizers seeking to optimize their financial gains without significantly modifying the opponent structure. Nonetheless, the consistent perspective of the tournament design literature (Groh et al. 2012; Horen and Riezman 1985; Vu and Shoham 2011) has been a focus

on the above three performance measures, hence our work is consistent with this literature.

Our work makes the following contributions. We propose two versions of our opponent choice design that addresses the above deficiencies in conventional tournament design. We describe an algorithm that computes, for each player, a sequence of opponents to choose that maximizes its tournament win probability. We study the performance of the opponent choice design using three established measures from the related literature. As shown by our computational study in Section 5, the opponent choice design outperforms the conventional design for the first two performance measures. For the third measure, the results closely match those from the conventional design. In all cases, our design achieves reasonable balance. We also study the robustness of our results by considering tournaments with incomplete information. Finally, we show that our opponent choice design reduces incentives for shirking under both individual and group strategic behavior, and provides several other practical advantages.

This paper is organized as follows. Section 2 lists some known applications of opponent choice in tournaments, and reviews the literature of tournament design. Section 3 defines the problem and provides examples. Section 4 describes a general tournament round, provides an algorithm by which any player can maximize its tournament win probability, and describes a large data set from men's professional tennis tournaments. In Section 5, we first provide a computational study of tournament win probabilities using the tennis data. We also study the sensitivity of the three performance measures discussed above to increasing irregularity in the players' pairwise win probability matrix. Section 6 discusses the performance of the opponent choice tournament design under two cases with incomplete information. In Section 7, we identify reductions in shirking behavior that the opponent choice tournament design provides, as well as limits to such reductions. Section 8 provides a conclusion and some suggestions for future work.

2 Previous work

In this section, we describe several tournaments where opponent choice is used in practice and review the most closely related literature of tournament design.

Several tournaments currently include opportunities for opponent choice:

- The Austrian League (hockeydb.com 2023).
- The Southern Professional Hockey League (U.S.) (Sports Illustrated 2018).

- The U.S. Bridge Federation (2019).
- The PRO Chess League (Chess.com 2017).

In addition, Inside Hook (2020) provides details of a preliminary proposal to expand the U.S. Major League Baseball playoffs to include additional teams, and permit higher ranked teams to choose their opponents. Despite the growing popularity of this tournament design in practice, the literature includes only one directly related study which we discuss below.

Since the research literature on tournaments is extensive, we focus on a few closely related works. A conventional single elimination tournament design that can be represented by a symmetric binary tree with fixed seeds is called a bracket, as used for example in regional play at the NCAA Men's and Women's basketball tournaments. However, for any tournament, the positioning of the variously ranked players allows for alternative bracket designs. This problem of bracket design, or "seeding problem", has been extensively discussed in the literature. In an early study of tournament design, Glenn (1960) compares all possible brackets with four players, to evaluate the probability that each player will win, and the expected number of games. Searls (1963) extends this analysis to eight players by comparing single and double elimination tournament designs, under single game and best-of-three-games formats. He finds that double elimination with best-of-three-games format provides the highest probability for the best player to win and the highest expected number of games.

Horen and Riezman (1985) compare different brackets for multiple round single elimination tournaments where the pairwise win probability matrix satisfies properties that enable the players to be ranked in a natural way, which we term a *medium ranking*. They apply four performance criteria: (i) does the bracket maximize the probability that the best player wins the tournament? (ii) Is it order preserving, i.e., no stronger player has a lower probability of winning the tournament than a weaker player? (iii) Does it maximize the probability that the best two teams meet in the final, and (iv) does it maximize the expected value of the winning team? For tournaments with four players, there are only three distinct brackets, and the conventional matchup of the strongest and weakest players at the semifinal round is the unique one that satisfies criteria (i) – (iv). For tournaments with eight players, there are 315 distinct brackets. Criterion (i) is satisfied by eight of them. However, for criterion (ii), no bracket is satisfactory.

Vu and Shoham (2011) make the stronger assumption that, for any pair of players, player P_i has a higher probability than player P_j of beating every other player. We denote

this situation as the existence of a *strong ranking*. Under this assumption, they study the effect of different brackets in general size tournaments, with a view to finding a seeding that achieves order preservation for every such pair of players. They show that, for a tournament with eight or more players, no bracket tournament design can guarantee this criterion for an arbitrary win probability matrix.

Groh et al. (2012) study an elimination tournament including heterogeneous players with known ability. Each match is studied as an auction, with equilibrium efforts characterized by mixed strategies. In any given match, win probabilities are influenced by the expected outcomes of other matches, which define expected future opponents. For tournaments with four players, it is possible to define optimal seedings to maximize the probability of the top player winning, and of the final being played between the top two teams. Brown and Minor (2014) consider a two-round tournament where the probability that a stronger player beats a weaker one is reduced by previous expenditure of effort (a "spillover effect") or by the expectation of playing a stronger future competitor at a later round (a "shadow effect"). With data from high-stakes betting tournaments, it is shown that these two effects influence match outcomes.

Vong (2017) considers the problem of strategic manipulation in multiple round tournaments. His definition of strategic manipulation is more general than shirking. Specifically, a tournament that is free of strategic manipulation is characterized as one where (a) full effort exists as a subgame perfect equilibrium, and (b) each such equilibrium coincides with the outcome from a full-effort social choice function. It is shown that allowing only the top-ranked player from each group to advance to the next round of the tournament is both necessary and sufficient to achieve these properties, under an arbitrary sorting rule that sorts qualifying players from one stage to the next. However, as Vong (2017) notes, eliminating all but one player from every group would quickly reduce interest in the tournament and disappoint spectators who have traveled far to watch the event, such as at the FIFA World Football Cup. This practical concern motivates our search for an alternative tournament design. By allowing higher ranked players to choose their opponents in the opponent choice design, it is typically better for a player to have higher rank, *ceteris paribus*, since then it has more available choices of opponent for the next round.

Guyon (2018) proposes the use of a *global ranking* at the elimination stage to remove group advantage, for tournaments where the number of groups is not a power of two. Under this ranking, all teams that are qualified for the elimination stage are ranked one after another based on

performance at the group stage. The proposed model is used by UEFA to modify the elimination bracket in the 2020 UEFA Euro Football Championship to minimize group advantage.

Guyon (2022) proposes a tournament design where the players are ranked based on their earlier performance. Then, in ranked order, the surviving players choose their opponents at the next round. He also considers a variant in which players in the top half of the ranking cannot be chosen by others. Within the context of a football tournament, the ranking is established first by points and then in the event of ties, by goal difference. Assuming that the tournament designer has the objective of maximizing the number of games that take place within the home country of a player, he applies three variations of this design to data for the 2020 UEFA Euro Football Championship. Our work, which is described below, differs from that of Guyon (2022) in several ways. First, we consider various levels of information about pairwise win probabilities. Second, we make the natural assumption that *each player* chooses its opponents to *maximize its tournament win probability*. Third, we evaluate the results of the opponent choice design against the three reasonableness criteria described by Horen and Riezman (1985). Fourth, we discuss the mitigation of shirking that is problematic in the conventional bracket design. A preliminary version of this work was presented as a Tutorial at the October 2022 INFORMS Annual Meeting.

3 Proposed design and examples

We consider a tournament consisting of a preliminary stage that establishes a *weak ranking* of players. Then, players compete in multiple rounds of single elimination plays until, at the last round, two players compete, and one player wins the tournament. Specifically, the single elimination stage consists of n rounds and $N = 2^n$ players. The players' weak ranking when they enter the single elimination stage is typically established through earlier performance, either over an entire season (e.g., the NFL and MLB regular seasons) or at the preliminary stage (e.g., the FIFA World Football Cup). For a bracket tournament design, this ranking completely specifies the matches that ensue for the winners at each round.

A weak ranking means that a higher-ranked player does not necessarily possess a greater than 0.5 win probability against a lower-ranked player. Further, a weak ranking does not imply that, for any pair of ranked players P_i and P_j ($i < j$), P_i has a higher probability than P_j of beating every other player in the tournament. In practice, the conditions on the win probability matrix considered by Horen and Riezman (1985) and Vu and Shoham (2011) are not commonly found

in tournaments where $N \geq 4$. The reason is the existence of anomalies for various pairs of players that prevent such a strong ranking. Nonetheless, we number the players in their weakly ranked order: $P_1, P_2, \dots, P_N$. Let p_{ij} denote the probability that P_i beats P_j in single elimination play. Unless otherwise stated, we assume that all the players know all such probabilities. Further, let $q_i^x, i = 1, \dots, N$ denote the probability that player P_i wins the tournament under a given tournament design x . If the tournament design under consideration is clear from context, we omit the superscript.

Our game model features opponent choice. We assume each player has accurate knowledge of all pairwise win probabilities. When choosing an opponent, each player maximizes its tournament win probability. Players choose their opponents sequentially, one after another. A player knows all the opponent choices that have been made when it makes its own choice. A player also takes other players' subsequent opponent choice decisions and its own choices in the later rounds into consideration when making its choice. Therefore, the opponent choice constitutes a multiple-stage Stackelberg game (Fudenberg and Tirole 1993).

We address the issue of ties for the weak ranking. Most U.S. major sports have detailed tie-breaking rules, based on earlier performance, that ensure a unique ranking. For example, the National Football League (NFL.com 2019), has 11 sequential performance-based rules for breaking ties between players, followed if necessary by a coin toss. The National Basketball Association (NBA.com 2019) uses a similarly elaborate tie-breaking system. Tournaments that form a single elimination bracket using very limited within-group performance information, for example the FIFA World Football Cup, could experience more ties for ranking. Tie-breaking options here include the use of a pre-tournament ranking, or of more detailed information, for example "goal difference" in football, from group performance.

We summarize the opponent choice tournament design as follows.

Opponent choice

1. Players compete in a preliminary stage, possibly lasting up to an entire season, that establishes a weak ranking among them.
2. Players compete in multiple rounds of single elimination play, by first choosing their opponent in weakly ranked order at each round.
3. When allowed to choose their next opponent, players do so in order to maximize their tournament win probability.

4. Players have full information, i.e., accurate knowledge of all pairwise win probabilities. This assumption is modified in Section 6.

As discussed in Section 1, one of the opponent choice design objectives is to motivate the players to perform well, including at the preliminary stage. However, we address this through *design changes at the single elimination stage*. Under a conventional bracket design for the single elimination stage, the first ranked player meets a specified opponent in single elimination. However, based on recent performance or other factors, that opponent may be perceived as a “bad matchup”, whereas another qualified player may be seen as easier to beat. This incentivizes shirking at the preliminary stage, in order to qualify but without being ranked first. The opponent choice design addresses this issue by allowing the first ranked player to choose its opponent, and similarly for other highly ranked players in ranked order.

Horen and Riezman (1985) compare different brackets for single elimination tournaments with four or eight players. They make several assumptions about the pairwise win probability matrix: (i) $0.5 \leq p_{ij} \leq 1$ for $1 \leq i < j \leq N$, (ii) $0 \leq p_{ij} \leq 0.5$ for $1 \leq j < i \leq N$, (iii) $p_{ij} + p_{ji} = 1$ for $1 \leq i, j \leq N$, and (iv) p_{ij} is nondecreasing in j for $1 \leq i, j \leq N$. These conditions imply that the players can be ranked such that each player ranks all the lower ranked players in the same sequence according to probability of beating them. If there exists an ordering of both the rows and columns of the pairwise win probability matrix such that conditions (i) – (iv) are satisfied, then the win probability matrix is said to be *strongly stochastically transitive* (David 1959), or SST.

The weak ranking we assume for the players when they enter the single elimination stage does not, in general, satisfy the SST conditions. In practice, win probability matrices are often not SST, especially for $N \geq 4$. As an example, we present data from the leading men’s professional tennis rivalries of the last 15 years, current as of September 2021 and as shown in Table 1, where the entries in row i and column j are the career wins and losses by the player in the row and the empirical estimate of p_{ij} based on those results.

It can be observed from Table 1 that the 4×4 submatrix involving the first four players – Novak Djokovic, Rafael

Nadal, Roger Federer and Andy Murray – is SST. However, the overall 5×5 matrix formed by the addition of Juan Martin Del Potro is not SST. This is because Djokovic has a better win record of 0.800 against Del Potro than his 0.694 record against Murray; whereas, Nadal has a better win record of 0.708 against Murray than his 0.647 record against Del Potro. In this sense, Del Potro is a “bad matchup” for Nadal, relative to their overall standards of play. Hence, no ordering of the players results in an SST matrix.

For a more extreme example of a bad matchup, consider the win probabilities in Table 2. A conventional bracket would match P_1 and P_4 in Table 2, to the disadvantage of player P_1 . In this case, the only strategy that enables P_1 to win the tournament is to play against P_2 , which enables P_3 to eliminate P_4 .

Remark 1. As shown by Horen and Riezman (1985), given an SST win probability matrix, player P_1 ’s bracket assignment to play against P_4 is optimal. It is in cases where the matrix contains non-SST irregularities that other alternatives become more attractive to P_1 .

We observe that, even for an SST win probability matrix with $N = 4$, the strategy of playing against P_3 may not beat the strategy of playing against P_2 . Consider the SST matrix in Table 3, where player P_1 ’s tournament win probability by choosing player P_2 is $0.5(0.5 \times 0.5 + 0.5 \times 1) = 0.375$ and by choosing player P_3 is $0.5(1 \times 0.5 + 0 \times 1) = 0.25$.

We develop further insight from a more general situation where one difficult player should be avoided, as shown in Table 4. Clearly, player P_1 cannot choose to play against player P_4 and win the tournament. If P_1 chooses P_2 , then $q_1 = p_{12}p_{34}p_{13}$. If P_1 chooses P_3 , then $q_1 = p_{13}p_{24}p_{12}$. Therefore, P_1 should choose P_2 if and only if $p_{12}p_{34}p_{13} \geq p_{13}p_{24}p_{12}$.

Table 2: Example of a bad matchup for player P_1 .

	P_2	P_3	P_4
P_1	1	1	0
P_2		0	0
P_3			1

Table 1: Win probability matrix for men’s professional tennis rivalries.

	R. Nadal	R. Federer	A. Murray	J.M. Del Potro
N. Djokovic	[30–28] 0.517	[27–23] 0.540	[25–11] 0.694	[16–4] 0.800
R. Nadal		[24–16] 0.600	[17–7] 0.708	[11–6] 0.647
R. Federer			[14–11] 0.560	[18–7] 0.720
A. Murray				[7–3] 0.700

Table 3: Example of SST matrix where playing against P_2 dominates playing against P_3 .

	P_2	P_3	P_4
P_1	0.5	0.5	1
P_2		0.5	1
P_3			0.5

Table 4: Example of one difficult opponent.

	P_2	P_3	P_4
P_1	p_{12}	p_{13}	0
P_2		p_{23}	p_{24}
P_3			p_{34}

or equivalently $p_{34} \geq p_{24}$. This result is intuitive, since P_1 chooses to let the other player, i.e., P_2 or P_3 , which has the better chance to eliminate P_4 do so.

Remark 2. Observe that player P_1 's choice of player P_2 or P_3 as opponent in Table 4 is independent of whether p_{12} or p_{13} is larger. Suppose $p_{12} = p_{13} + \epsilon$, where $\epsilon > 0$ is small, but $p_{24} \gg p_{34}$. Then, player P_1 should play against P_3 , even though it is easier to beat P_2 . This shows by example that player P_1 's strategy of choosing the player, which it beats with highest probability, does not necessarily maximize its tournament win probability, even for $N = 4$.

4 Tournament design and algorithm

Section 4.1 describes the opponent choice design for a single elimination tournament of general size. In Section 4.2, we describe an algorithm to determine the tournament win probability of every player, based on their ranking, under both static and dynamic designs. Section 4.3 describes a data set from men's professional tennis. The Appendix provides a numerical example of the algorithm for a tournament with $N = 8$ players.

4.1 Design

We consider a generic single elimination stage of a tournament with n rounds and $N = 2^n$, $n = 2, 3, \dots$, players. The players have a weak ranking, i.e., a *total order*, entering the single elimination stage. In total, in the first round $N/2 - 1$ players choose their opponents sequentially, constituting an $(N/2 - 1)$ -stage Stackelberg game. Alternative designs where fewer than $N/2 - 1$ players freely choose their opponents can be described by our model and solved by our algorithm as special cases.

After the $N/2$ first round games, the $N/2$ winners enter the second round. There are two alternative models for how the ranking of players is affected by earlier results. This ranking is important, since it determines the sequence in which players choose their opponents in the next round. We consider both static and dynamic rankings in our work.

More generally, various ranking rules can be adopted, for example interpolating between the two models we consider, as in the Elo system (Chess.com 2022) used to determine chess ratings.

Next, we provide an explanation of dynamic ranking. For simplicity, we assume that the $N/2$ games in the first round are numbered as games $1, 2, \dots, N/2$, in the order of the ranking of the higher ranked player in each game. At the second round, the winner of each game inherits the number of the previous game they won, and hence the $N/2$ winners are reordered as $P_{(1)}, P_{(2)}, \dots, P_{(N/2)}$ accordingly, where $(1), (2), \dots, (N/2)$ are the players' immediately previous rankings. Accordingly, the highest ranked $N/4 - 1$ of the $N/2$ winners choose their opponents in ranked order, following the same procedure as in the first round. For example, suppose that at some round, player P_8 beats player P_1 in the first game, and player P_5 beats player P_2 in the second game. Then, at the next round, player P_8 has ranking 1 and chooses its opponent first, and player P_5 has ranking 2 and chooses second. The rounds continue similarly, until a tournament winner is determined.

At round h , 2^{n-h+1} players play 2^{n-h} games and $2^{n-h} - 1$ players can choose their opponents. We assume that each player has full information about the pairwise win probabilities, and if able to choose its next round opponent, does so to maximize its probability of winning the tournament.

4.2 Algorithm

We now describe our algorithm to determine each player's choice of its next round opponent. The algorithm works under both static and dynamic ranking.

The opponent choice problem is solved using a backward dynamic programming recursion. The starting point is the final round, where each player's opponent is fixed and its win probability is known, for every possible pair of finalists. The algorithm uses this information to determine optimal opponent choices for the semifinal round, quarter-final round, and so on. We describe how to solve the first round problem. To do so, we assume that the second round problem with $N/2$ players has been solved. That is, for any given $N/2$ players with known pairwise win probabilities, we have found the best choices for the $N/4 - 1$ players which may choose their opponents. We further assume that in the first round, a subset of players have had their opponents decided, either by their own choice or as a result of being chosen by higher ranked players. Then, among the players with opponents not yet decided, the highest ranked player chooses an opponent, with the objective of maximizing its tournament win probability. Under these assumptions, we now describe our dynamic programming

algorithm for a generic n round tournament with $N = 2^n$ players.

4.2.1 Algorithm opponent choice

Input

p_{ij} , for $i, j = 1, \dots, N, N = 2^n, i \neq j$. Let S_a denote the set of all the players.

Value Function

Consider a given set S of players with relative rankings $1, \dots, |S|$ and respective original rankings $(1), \dots, (|S|)$. Assume that the players in S are all the players in a tournament round and make optimal opponent choices. Let $Q(S) \equiv \{q_{S,(1)}, \dots, q_{S,(|S|)}\}$ be the corresponding tournament win probabilities of players in S .

Boundary Condition

When $|S| = 2$, i.e., only two players with relative rankings 1 and 2, we have $q_{S,(1)} = p_{(1)(2)}$ and $q_{S,(2)} = p_{(2)(1)}$, where (1) and (2) are the two players' original rankings. Hence we assume that, for any $S \subseteq S_a, |S| = 2, 4, \dots, N/2$, we have found $Q(S) = \{q_{S,(1)}, \dots, q_{S,(|S|)}\}$.

Further, when $S = S_a$ and the opponent of each player is decided, there are $2^{N/2}$ possible sets of the $N/2$ winners. Define ordered sets $\mathbf{X}$ and $\mathbf{X}'$ such that players in set $\mathbf{X}$ play against players in set $\mathbf{X}'$ in the first round, in the same order as in their respective sets, where $|\mathbf{X}| = |\mathbf{X}'| = N/2$. Let $\Omega(\mathbf{X}, \mathbf{X}')$ denote the collection of all possible sets of $N/2$ winners, and p_S denote the probability of the occurrence of set $S \in \Omega(\mathbf{X}, \mathbf{X}')$. Let x_i and x'_i be the initial rankings of the i th players in $\mathbf{X}$ and $\mathbf{X}'$, respectively. We have $p_S = \prod_{i=1}^{N/2} p_{x_i x'_i}$. Then, the tournament win probability $q_j = q_{S_a,j}$ of player $P_j, j = 1, \dots, N$, is:

$$q_{S_a,j} = \sum_{S \in \Omega(\mathbf{X}, \mathbf{X}')} p_S q_{S,j}. \quad (1)$$

Optimal Solution Value

$Q(S_a)$.

Recurrence Relation

At any round, the algorithm works by evaluating all subsets of smaller size before all subsets of larger size. When $N - 2$ players have their opponents already decided and two players P_i and P_j , where $i < j$, do not, player P_i will play player P_j as its opponent, for any $1 \leq i < j \leq N$. Next, the algorithm evaluates all subsets of four players, using the previously computed information for any pair of last two players, and so on, until the optimal choices of all N players and their win probabilities are determined. As in the boundary condition, we define ordered sets $\mathbf{X}$ and $\mathbf{X}'$ with $|\mathbf{X}| = |\mathbf{X}'|$, such that players in set $\mathbf{X}$ play against players in set $\mathbf{X}'$, in the same order as in their respective sets. By assumption,

the collection of all possible sets of $N/2$ winners $\Omega(\mathbf{X}, \mathbf{X}')$ is known for $|\mathbf{X}| = |\mathbf{X}'| \geq m$, for some $1 \leq m \leq N/2$. Next, we assume that $2m - 2$ players have their opponents decided in ordered sets $\mathbf{X}$ and $\mathbf{X}'$ with $|\mathbf{X}| = |\mathbf{X}'| = m - 1$. Let set $\mathcal{U}$ contain all the players with undecided opponents, where player P_j has the highest ranking in $\mathcal{U}$ and $\mathbf{X} \cup \mathbf{X}' \cup \mathcal{U} = S_a$. Then, player P_j chooses its opponent to maximize its tournament win probability $q_j = q_{S_a,j}$, as follows:

$$q_{S_a,j} = \max_{i \in \mathcal{U}} \sum_{S \in \Omega(\mathbf{X} \cup \{j\}, \mathbf{X}' \cup \{i\})} p_S q_{S,j}, \quad (2)$$

where $\Omega(\mathbf{X} \cup \{j\}, \mathbf{X}' \cup \{i\})$ is known by assumption, since $|\mathbf{X} \cup \{j\}| = |\mathbf{X}' \cup \{i\}| = m$. In (2), collection $\Omega(\mathbf{X}, \mathbf{X}')$ is defined recursively.

Theorem 1. *Algorithm Opponent Choice finds the optimal sequence of opponent choices for all $N = 2^n$ players, and their tournament win probabilities. Under the dynamic ranking, the algorithm's time complexity has order*

$$\sum_{h=1}^{n-1} \frac{(2^n - 2^{n-h+1} + 1)^{2^{n-h+1}} (2^{n-h+1})!}{(2^{n-h})!}. \quad (3)$$

Under the static ranking, the algorithm's time complexity has order

$$(2^n)! \sum_{h=1}^{n-1} \frac{1}{(2^{n-h})! (2^n - 2^{n-h+1})!}. \quad (4)$$

Proof. From the recurrence relation, the algorithm:

- (i) considers all possible transitions from each round to the previous round, using Equation (1), and
- (ii) considers all possible transitions at each round, from the case where $2m$ players have decided opponents to the case where $2m - 2$ players have decided opponents using Equation (2), for all relevant values of m .

Therefore, the algorithm compares the tournament win probabilities of all possible states, and finds an optimal solution.

For the time complexity under dynamic ranking, note that at any round, a player with an initial rank of k , for any $1 \leq k \leq 2^n$, can only be ranked $1, \dots, k$. Thus, at any round h with $N/2^{h-1} = 2^{n-h+1}$ players, the last-ranked player must have an initial rank no higher than 2^{n-h+1} , and there are $2^n - 2^{n-h+1} + 1$ possible players to fill the rank. Once the last ranked player is given, the second-last-ranked player must have an initial rank of no higher than $2^{n-h+1} - 1$, and there are $2^n - 2^{n-h+1} + 1$ remaining possible players to fill the rank. Continuing this backward process, we see that for every rank at round h , there are $2^n - 2^{n-h+1} + 1$ players to

fill. Therefore, the number of possible sequences of 2^{n-h+1} players at round h is $(2^n - 2^{n-h+1} + 1)2^{n-h+1}$.

At the h th round with 2^{n-h+1} players, the player with the first choice chooses from $2^{n-h+1} - 1$ players, the player with the second choice chooses from $2^{n-h+1} - 3$ players, and so on. Hence, there are $(2^{n-h+1} - 1)!!$ possible choices. For each choice, there are 2^{n-h} games with $2^{2^{n-h}}$ possible outcomes. Therefore, assuming that optimal opponent choices have been obtained for all sequences of players at the $(h+1)$ th round, with a backward recursion, we can compute an optimal opponent choice for each sequence at the h th round by $2^{2^{n-h}}(2^{n-h+1} - 1)!! = 2^{2^{n-h}}(2^{n-h+1})!/(2^{n-h+1})!! = 2^{2^{n-h}}(2^{n-h+1})!/(2^{2^{n-h}}(2^{n-h})!) = (2^{n-h+1})!/(2^{n-h})!$ enumerations. Therefore, the total number of enumerations at the h th round is

$$(2^n - 2^{n-h+1} + 1)2^{n-h+1} \frac{(2^{n-h+1})!}{(2^{n-h})!}. \quad (5)$$

Summing (5) up for $h = 1, 2, \dots, n-1$ yields (3).

Under the static ranking, note that the ranking is uniquely determined for every subset of players. At the h th round, we only need to consider subsets of 2^{n-h+1} players. Specifically, there are a total of $2^n C_{2^{n-h+1}}$ subsets. Consequently, we can replace (5) by

$$2^n C_{2^{n-h+1}} \frac{(2^{n-h+1})!}{(2^{n-h})!} = \frac{(2^n)!}{(2^{n-h})!(2^n - 2^{n-h+1})!}. \quad (6)$$

Summing (6) up for $h = 1, 2, \dots, n-1$ yields (4). $\square$

For $n = 3$, i.e., 8 players, the values of (3) and (4) are 9.18×10^3 and 2.52×10^3 , respectively. For $n = 4$, i.e., 16 players, the values of (3) and (4) are 7.28×10^{10} and 5.41×10^8 , respectively. A personal computer can solve such instances optimally within an hour. Therefore, the opponent choice design could easily be used, for example, for the single elimination stage of the FIFA World Football Cup that includes 16 players.

4.3 Data

In order to test the opponent choice design on real sports tournaments, we access data from Sports Hub Data (2022a, b) for Association of Tennis Professionals (ATP) men's tennis tournaments from 1991 to 2017. This data set includes 99,189 tennis matches over 2215 tournaments played during that period.

To focus on matches where players seeded from No. 1 to No. 16 potentially play each other, we restrict the data set to the 28,577 matches that were played at the round-of-16 or a later round in those tournaments. Such potential matches actually occur in 6195 matches, which are won by

the higher (respectively, lower) seeded player in 3914 (resp., 2281) instances. We use the results of these 6195 matches to estimate win probabilities between all pairs of players seeded between No. 1 and No. 16, based on win frequency. For example, there are 177 matches between the No. 1 and No. 2 seeds, with the No. 1 seed winning 97 of them, thus giving an empirically estimated win probability for the No. 1 seed of $97/177 = 0.548$. In case the number of matches between two seeds is very small, we add the same number to the number of wins of each player to ensure that the number of wins by each seed is not less than 5. For example, there are 2 matches between the No. 15 and No. 16 seeds with the No. 15 seed winning both of them; after adjustment, we have the No. 15 seed winning 7 out of 12 matches between the two seeds. The use of the number 5 in this context represents an intuitive Bayesian approach aimed at balancing two considerations: (a) the restraint from exerting substantial alterations upon the dataset and (b) the intention to make corrections for an exceedingly small sample size, which is inherently unreliable. We note that the rigorous assessment of the precision of the estimates presented in Table 5 deserves in-depth investigation.

The pairwise win probability matrix is shown in Table 5, with values rounded to three decimal places. Note that the data of each numbered seed is collected across different players with the same seeding at different matches. Therefore, each seeded player is a generic player whose win probabilities are also against other generic players with the specified seedings. Since we have an extensive data set, the obtained win probability matrix in Table 5 is representative.

5 Computational study

In this section, we use computational studies to compare the performance of the opponent choice design, under both static and dynamic rankings, to that under a conventional bracket design. Specifically, we evaluate whether the opponent choice design (a) provides a reasonably higher win probability to the top ranked player(s) than a bracket, (b) provides a reasonably higher probability of the top two players meeting than a bracket, and (c) provides a ranking by tournament win probabilities that closely matches the original ranking of the players. Section 5.1 computes the tournament win probabilities using the opponent choice design for generic seeded tennis players at the semifinal and quarterfinal rounds and round-of-16, using the data set of Section 4.3, and compares them with the same probabilities using a conventional bracket design. In Section 5.2, we compare the effect of increasing irregularity in a pairwise win

Table 5: Win probability matrix for seeded tennis players.

	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	0.548	0.649	0.663	0.727	0.708	0.764	0.721	0.763	0.795	0.783	0.813	0.806	0.886	0.788	0.761
2		0.487	0.687	0.632	0.620	0.600	0.668	0.784	0.634	0.619	0.848	0.725	0.833	0.803	0.754
3			0.431	0.567	0.613	0.625	0.571	0.609	0.438	0.645	0.548	0.714	0.590	0.750	0.645
4				0.520	0.508	0.667	0.653	0.563	0.773	0.412	0.571	0.739	0.607	0.600	0.821
5					0.500	0.607	0.327	0.500	0.667	0.607	0.643	0.706	0.500	0.667	0.500
6						0.536	0.558	0.455	0.559	0.596	0.643	0.500	0.560	0.643	0.722
7							0.409	0.583	0.607	0.579	0.565	0.385	0.583	0.500	0.615
8								0.483	0.581	0.600	0.406	0.385	0.417	0.545	0.706
9									0.583	0.455	0.615	0.545	0.545	0.643	0.385
10										0.545	0.706	0.357	0.545	0.545	0.583
11											0.500	0.583	0.583	0.333	0.385
12												0.615	0.667	0.667	0.385
13													0.500	0.545	0.385
14														0.500	0.500
15															0.583

probability matrix on the performance measures (a), (b) and (c) described above.

5.1 Win probabilities by seeding

Using the seeding of the tennis players as the ranking, we compute the tournament win probabilities of the various players at the semifinal and quarterfinal rounds and round-of-16, as shown in Tables 6–8, respectively. These probabilities under opponent choice are computed by assuming that the highest ranked player(s) select the opponent(s) that maximize their tournament win probability. For example, at the semifinal round, the No. 1 seed has a tournament win probability of 0.398 if choosing to play the No. 4 seed, compared to 0.360 against the No. 2 seed, and 0.379 against the No. 3 seed; it therefore chooses to play the No. 4 seed. The other semifinal match is then between the Nos. 2 and 3 seeds. In Table 6, the various players have identical tournament win probabilities for the bracket design and the opponent choice design.

Table 7 contains quarterfinal results, where q_i^D and q_i^S are the tournament win probabilities of player i under

Table 7: Tournament win probabilities for seeded tennis players at quarterfinal round.

Seed i	1	2	3	4	5	6	7	8
Opponent D	7	8	6	5	4	3	1	2
q_i^D	0.339	0.202	0.139	0.098	0.063	0.066	0.038	0.055
Opponent S	7	4	6	2	8	3	1	5
q_i^S	0.344	0.203	0.141	0.081	0.045	0.067	0.037	0.082
Opponent B	8	7	6	5	4	3	2	1
q_i^B	0.312	0.185	0.140	0.115	0.076	0.073	0.044	0.056

Table 8: Tournament win probabilities for seeded tennis players at round-of-16.

Seed i	1	2	3	4	5	6	7	8
Opponent	14	12	15	16	13	11	10	9
q_i^D	0.333	0.196	0.110	0.085	0.057	0.047	0.031	0.035
q_i^S	0.332	0.208	0.111	0.089	0.056	0.048	0.030	0.036
q_i^B	0.307	0.203	0.091	0.085	0.057	0.051	0.035	0.036
Seed i	9	10	11	12	13	14	15	16
Opponent	8	7	6	2	5	1	3	4
q_i^D	0.024	0.016	0.021	0.008	0.011	0.009	0.009	0.008
q_i^S	0.022	0.014	0.020	0.007	0.009	0.005	0.007	0.007
q_i^B	0.026	0.017	0.022	0.013	0.013	0.011	0.012	0.020

Table 6: Tournament win probabilities for seeded tennis players at semifinal round.

Seed i	1	2	3	4
Opponent i	4	3	2	1
q_i	0.398	0.259	0.194	0.150

dynamic and static rankings of players, respectively. For the opponent choice design, each player's quarterfinal round opponent is also shown, in row Opponent D under dynamic ranking and in row Opponent S under static ranking. Further, q_i^B is the tournament win probability of player i using

the conventional quarterfinal round bracket (1, 8, 4, 5, 3, 6, 2, 7). We discuss the results in Table 7. For this instance, seeds Nos. 1 and 2 both gain slightly from the opponent choice design, from being able to choose their opponents. In reality, the differences between top professional players are usually quite minor, and it is impractical to improve the winning odds of one top player by a large amount without compromising balance. Most other seeds have a lower tournament win probability, due to their limited choices and the chance of being chosen by a higher ranked player that does not occur at the quarterfinal round under a conventional bracket design. Seed No. 8 gains under the opponent choice design with static ranking because seed No. 5 is its quarterfinal opponent.

Table 8 contains round-of-16 results in a similar format. For this instance, using the opponent choice design, each player's round-of-16 opponents are the same under dynamic and static rankings of players. The main winner from the opponent choice design is seed No. 1, due to its first-mover advantage. Besides, seed No. 3 benefits from the opponent choice design with dynamic ranking, and seeds Nos. 2, 3, and 4 benefit from the opponent choice design with static ranking. The benefit they receive is paid for quite evenly by seeds Nos. 6 through 16. A particularly interesting case is seed No. 14, which is chosen by seed No. 1. This difficult matchup reduces its tournament win probability from 0.011 using the bracket design to 0.005 using the opponent choice design under the static ranking, but only to 0.009 under the dynamic ranking. The difference in these last two numbers results from the possibility that, by beating seed No. 1, seed No. 14 achieves the No. 1 ranking at the quarterfinal round under the dynamic ranking.

5.2 Sensitivity to matrix irregularity

We compare the effect of increasing but known irregularity in the pairwise win probability matrix on (a) the tournament win probabilities of the various players, (b) the probability that players P_1 and P_2 meet, and (c) order preservation relative to ranking, using the opponent choice design under both static and dynamic rankings and a conventional bracket design, for both the semifinal and quarterfinal rounds.

We begin with a regular SST matrix with $N = 2^n$ players where $p_{ij} = 0.5 + 0.05(j - i)$, for $1 \leq i < j \leq N$. We control the degree of irregularity δ by adding to each matrix entry a randomly generated value $d - U[-\delta, \delta]$, for $\delta = 0, 0.01, 0.02, \dots, 0.1$. We generate the value of d independently for each matrix entry. For the semifinal and quarterfinal rounds with $\delta > 0$, we have 6 and 28 random variables, respectively, and hence use a sample size of 10,000 for

each game. For the semifinal round the dynamic and static rankings are the same under the opponent choice design since only player P_1 can choose its opponent, whereas for the quarterfinal round they are different.

5.2.1 Tournament win probabilities

We consider the effect of increasing irregularity on the tournament win probabilities of players with various rankings. In Figures 1 and 2, the horizontal axis represents the degree of irregularity δ , and the vertical axis represents the mean tournament win probability of a specific player. For ease of presentation, the subgraphs show different ranges on the vertical axis. In Figure 1, the solid and dotted lines represent tournament win probabilities using the opponent choice design (qD, i) and using a conventional bracket design (qB, i), respectively. In Figure 2, the solid and dashed lines represent tournament win probabilities using the opponent choice design under the dynamic (qD, i) and static (qS, i) rankings, respectively, and the dotted line represents tournament win probabilities using a conventional bracket design (qB, i).

We discuss the semifinal round results shown in Figure 1. For the initial case, when $\delta \leq 0.025$, the win probability matrix is SST, and thus the conventional bracket design maximizes the tournament win probability of player P_1 (Horen and Riezman 1985) and performs the same as the opponent choice design for all the players. Using both designs, the tournament win probabilities of players P_1 and P_2 decrease as matrix irregularity increases, while those of players P_3 and P_4 increase with matrix irregularity. This is intuitive, since greater matrix irregularity gives lower ranked players a higher probability of winning a game. The opponent choice design benefits player P_1 the most relative to the conventional bracket design when matrix irregularity is high, since the design gives player P_1 an opportunity to choose an opponent besides player P_4 . Player P_2 also benefits slightly from the opponent choice design when the matrix irregularity is moderate. This is because, with moderate irregularity, player P_1 is more likely to choose player P_3 , and consequently player P_2 plays P_4 , which is on average better for player P_2 . However, when matrix irregularity is high, player P_2 may be chosen by player P_1 , and as a result its tournament win probability is slightly less than that using the conventional bracket. Player P_3 's tournament win probability is reduced by the opponent choice design, due to the probability that it is chosen by player P_1 . Finally, the opponent choice design favors player P_4 slightly, since its probability of being chosen by player P_1 is reduced.

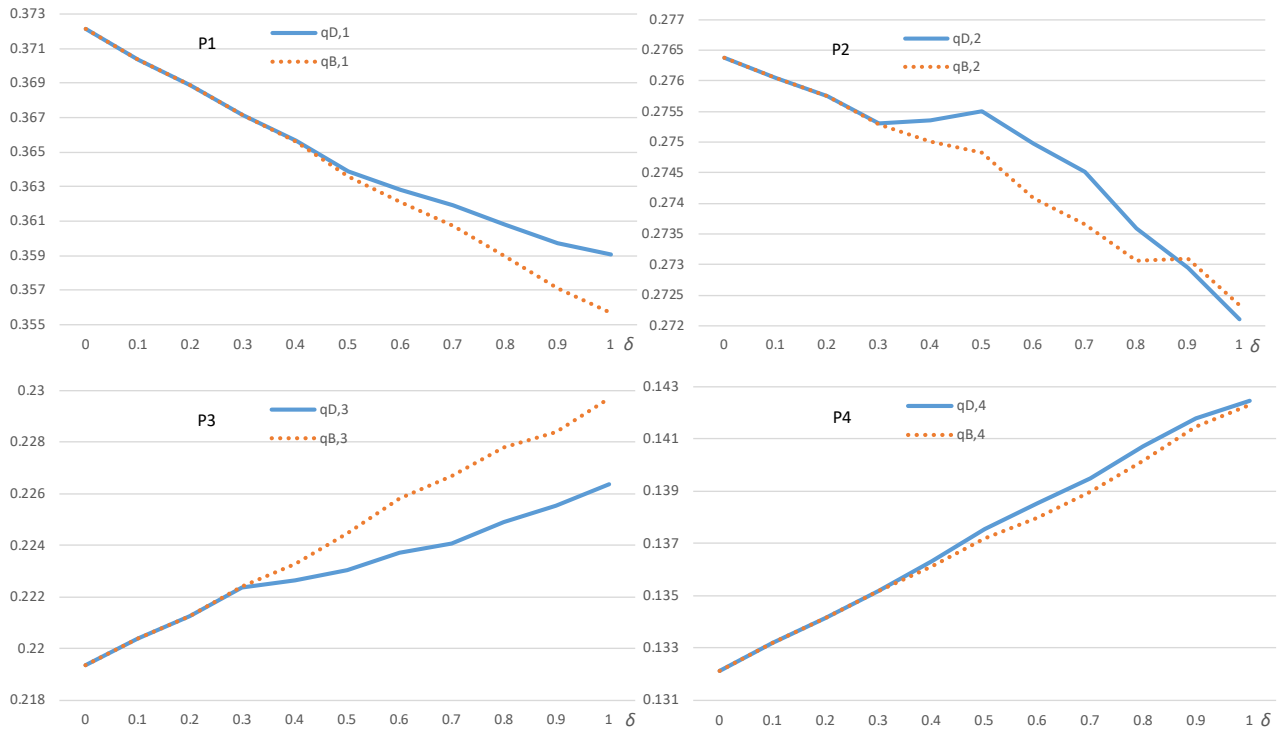


Figure 1: Tournament win probabilities for semifinal round.

We discuss the quarterfinal round results shown in Figure 2. The opponent choice design benefits player P_1 similarly under the dynamic and static rankings, and the two lines are indistinguishable in Figure 2. Moreover, the relative benefit is highest when matrix irregularity is high, since it is then that the value of choosing an opponent other than P_8 is the greatest. Suppose player P_1 loses to a player A and player P_2 wins at the quarterfinal round. Then, at the semifinal round under the static ranking player P_2 chooses its opponent first. However, under the dynamic ranking, player A , which is likely to be a weak player since it was chosen by player P_1 , chooses its opponent first. Hence, player P_2 benefits greatly from the flexibility of choice under the static ranking, but less under the dynamic ranking. The results for player P_8 are opposite to those for player P_2 . Players P_3 and P_4 's win probabilities are reduced under the opponent choice design, because they may face stronger opponents after the choice(s) of players P_1 and P_2 . Since player P_4 is unlikely to choose at the quarterfinal round, it benefits from the chance to improve its ranking dynamically. Finally, using the opponent choice design under the dynamic ranking, each player may have an opportunity to choose its opponents. But under the static ranking, players P_4 and P_5 have limited opportunity, and players $P_6 - P_8$ have no opportunity to choose opponents, and they are subject to

the choices of higher ranked players. Consequently, players $P_4 - P_8$ have consistently lower tournament win probabilities under the static ranking than under the dynamic ranking.

5.2.2 Top two players meeting

We compare the probabilities for players P_1 and P_2 to meet under different designs. Under the conventional bracket, players P_1 and P_2 can only meet in the final, whereas under the opponent choice design they can meet in any round. Our instances contain no cases where players P_1 and P_2 meet in the quarterfinal round, hence we report the probabilities for them to meet in the semifinal and final rounds. In Table 9, columns $q_{D4,12}$, $q'_{D4,12}$ and $q_{B4,12}$ contain the mean probabilities that players P_1 and P_2 meet in the final and semifinal rounds under the opponent choice design, and in the final under the conventional bracket design, respectively, for the semifinal round. Similarly, the next five columns are for the quarterfinal round, with the dynamic ($q_{D8,12}$, $q'_{D8,12}$) and static ($q_{S8,12}$, $q'_{S8,12}$) rankings separated. We observe that with small matrix irregularity $\delta \leq 0.03$, the designs perform similarly. But with increased irregularity, the opponent choice design achieves relatively greater probability for players P_1 and P_2 to meet. In the semifinal round, when irregularity is very high ($\delta \geq 0.09$), the chance

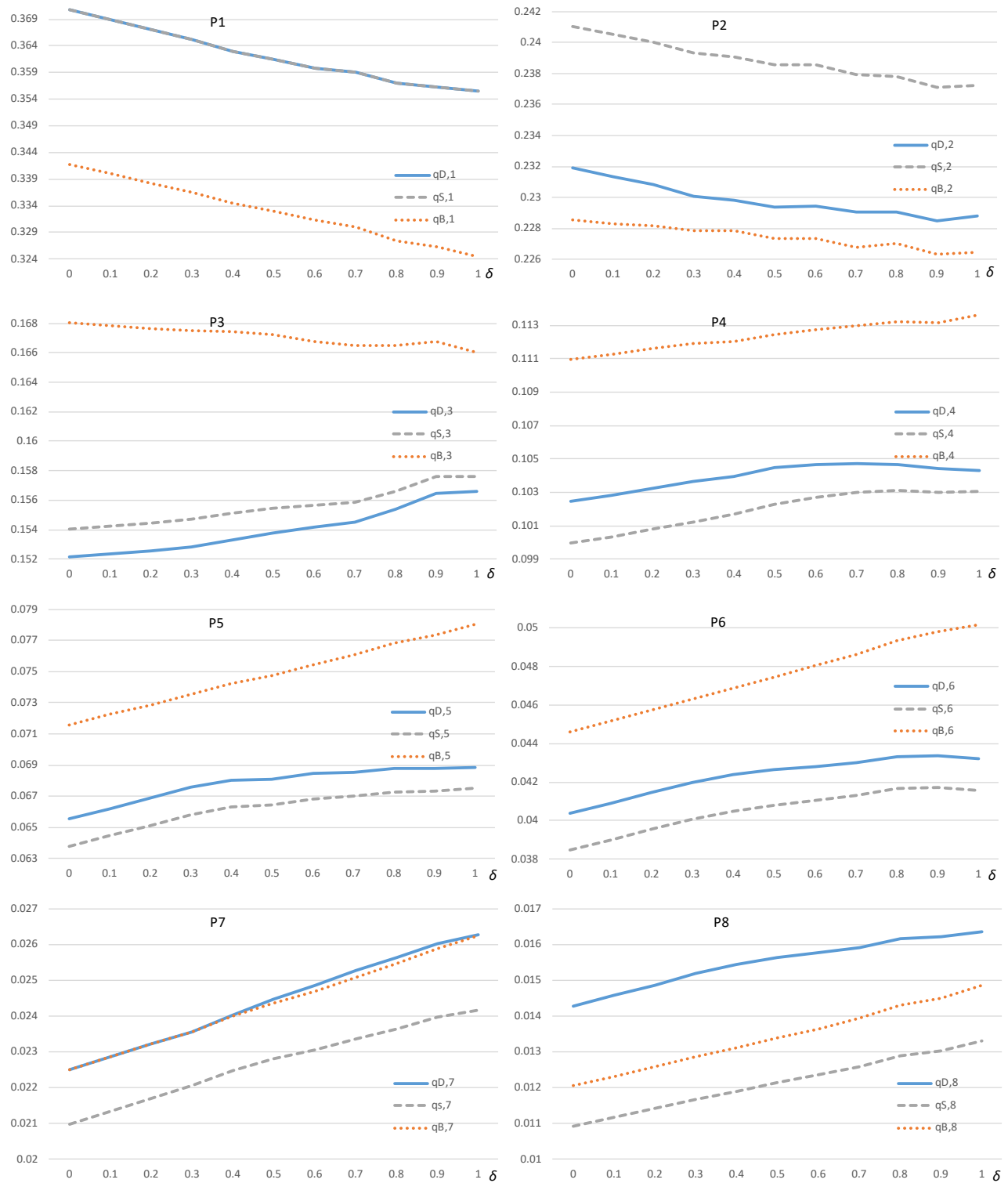


Figure 2: Tournament win probabilities for quarterfinal round.

for these players to meet in the final under the opponent choice design is slightly less, but is more than compensated by the probability that they meet in the semifinal round.

However, in the quarterfinal round, even with high irregularity and increased probability for players P_1 and P_2 to meet in the semifinal round, the opponent choice design still

Table 9: Probability of player P_1 meeting player P_2 .

Parameter	$q_{D4,12}$	$q'_{D4,12}$	$q_{B4,12}$	$q_{D8,12}$	$q'_{D8,12}$	$q_{S8,12}$	$q'_{S8,12}$	$q_{B8,12}$
$\delta = 0.00$	0.358	0.000	0.358	0.257	0.000	0.257	0.000	0.258
$\delta = 0.01$	0.356	0.000	0.356	0.256	0.000	0.256	0.000	0.256
$\delta = 0.02$	0.354	0.000	0.354	0.254	0.000	0.254	0.000	0.254
$\delta = 0.03$	0.353	0.000	0.352	0.252	0.000	0.252	0.000	0.252
$\delta = 0.04$	0.352	0.000	0.351	0.250	0.000	0.250	0.000	0.251
$\delta = 0.05$	0.352	0.000	0.349	0.250	0.000	0.250	0.000	0.249
$\delta = 0.06$	0.351	0.002	0.348	0.249	0.001	0.249	0.001	0.247
$\delta = 0.07$	0.348	0.013	0.346	0.249	0.003	0.249	0.003	0.245
$\delta = 0.08$	0.345	0.022	0.344	0.248	0.007	0.248	0.007	0.243
$\delta = 0.09$	0.339	0.039	0.343	0.247	0.011	0.247	0.011	0.241
$\delta = 0.10$	0.337	0.049	0.341	0.247	0.015	0.247	0.015	0.240
Mean	0.349	0.011	0.349	0.251	0.003	0.251	0.003	0.249

offers greater probability for them to meet in the final than the conventional bracket design does.

5.2.3 Rank preservation

We examine how the ranking of players by their tournament win probability under each design preserves their original ranking. For this performance measure, the bracket design performs well, but we show that the opponent choice design produces very similar results. For each design and δ value, we compute the *Kendall tau rank correlation coefficient* (Kendall 1938). Let $(x_1, y_1), \dots, (x_N, y_N)$ be a set of different rankings. A pair of rankings (x_i, y_i) and (x_j, y_j) are *concordant* if $x_i > x_j$ and $y_i > y_j$, or $x_i < x_j$ and $y_i < y_j$; and are *discordant* if $x_i > x_j$ and $y_i < y_j$, or $x_i < x_j$ and $y_i > y_j$. Specifically, the Kendall coefficient τ , where $-1 \leq \tau \leq 1$, is defined as

$$\tau = \frac{2(\text{number of concordant pairs} - \text{number of discordant pairs})}{n(n-1)}.$$

The results are summarized in Table 10. For the semifinal round, columns τ_{D4} and τ_{B4} represent the Kendall coefficients between the players' original ranking and their ranking by tournament win probability, as found by the opponent choice design and the conventional bracket, respectively. For the quarterfinal round, columns τ_{D8} , τ_{S8} and τ_{B8} represent the Kendall coefficients between the players' original ranking and their ranking by tournament win probability found by the opponent choice design with dynamic and static rankings, and the conventional bracket, respectively. The rows correspond to different values of the irregularity parameter δ . When $\delta \leq 0.03$, all the coefficients are 1.000. This confirms that, for these randomly generated win

Table 10: Kendall coefficient for player rankings.

Parameter	τ_{D4}	τ_{B4}	τ_{D8}	τ_{S8}	τ_{B8}
$\delta = 0.04$	0.998	0.998	1.000	1.000	1.000
$\delta = 0.05$	0.990	0.989	0.999	0.999	1.000
$\delta = 0.06$	0.970	0.970	0.996	0.997	0.999
$\delta = 0.07$	0.945	0.947	0.991	0.991	0.996
$\delta = 0.08$	0.921	0.924	0.983	0.983	0.990
$\delta = 0.09$	0.894	0.900	0.974	0.974	0.983
$\delta = 0.10$	0.870	0.876	0.964	0.964	0.973
Mean	0.941	0.943	0.987	0.987	0.991

probability matrices, the opponent choice design is perfectly order preserving. Table 10 starts with $\delta = 0.04$. The opponent choice design produces tournament win probabilities with similar rankings by dynamic and static ranking. With increasing irregularity δ , the opponent choice design achieves tournament win probabilities that are slightly less consistent with the players' ranking than those found by the conventional bracket. This is because the conventional bracket determines opponents strictly by ranking, hence every position in the ranking affects a player's opponent in the tournament. Whereas, in the opponent choice design only player P_1 chooses its opponent in the semifinal round, and only three players with higher seeding choose their opponents in the quarterfinal round. However, with increasing irregularity, the players' original ranking becomes a less accurate measure of their relative strength, hence order preservation with ranking becomes less equivalent to order preservation with player strength and a less relevant performance measure.

6 Incomplete information

In this section, we drop the above assumption of complete information. Section 6.1 considers a case where players have imprecise information about pairwise win probabilities. In Section 6.2, they have no information about games between other players.

6.1 Imprecise information

With information that is known to be imprecise, the players may simplify their estimates into a few categories; such as “win”, “draw”, and “lose”. Consider the following model for probability estimates. If the win probability is estimated as greater than 0.67, the player uses a value of 1; if the win probability is between 0.33 and 0.67, the player uses a value of 0.5; and if the win probability is smaller than 0.33, the player uses a value of 0. We study the tournament win probabilities

of the various players under incomplete information, using the opponent choice design under dynamic ranking at the quarterfinal round. The exact but unknown win probability matrix is generated as in Section 5.2. Due to the limited number of pairwise win probabilities, ties for tournament win probability may occur in opponent choice. We assume that each player breaks ties by choosing a player with the lowest ranking. Following the opponent choice of players using imprecise information, we compute the true tournament win probabilities of players using the matrix information, and compare the probabilities with those obtained by the opponent choice design under both dynamic ranking and the conventional bracket design with complete information.

The results are shown in Figure 3. As in Figure 2, the solid line represents tournament win probabilities using the opponent choice design under the dynamic ranking (qD, i), and the dotted line represents tournament win probabilities using the conventional bracket design (qB, i), both with complete information. The dashed line represents tournament win probabilities for opponent choice using the opponent choice design under the dynamic ranking with imprecise information (qE, i). The results show that, for all players except for P_2 and P_7 , the tournament win probabilities under opponent choice with incomplete information fall between those obtained by the other two approaches with complete information. This is intuitive: with simplified estimates of 0, 0.5, and 1 for the pairwise win probabilities, players cannot accurately differentiate the other players in their opponent choice and hence make decisions that are more similar to those under the conventional bracket design. However, players P_2 and P_7 benefit from opponent choice compared with the conventional design, even more with incomplete information than with complete information. This is because with imprecise information, the first mover player P_1 , cannot make opponent choice decisions that are as accurate as in the case of complete information. Hence, the second mover player P_2 can improve its tournament win probability. Player P_7 benefits from opponent choice with imprecise information since it is less likely to be chosen by player P_1 than under opponent choice with full information and by player P_2 than under the conventional bracket design. We also note that player P_2 's tournament win probability is not monotonic in the level of matrix irregularity δ under imprecise information. This is because, when $\delta = 0$, the estimated pairwise win probabilities are close to their true values.

6.2 No information

A situation with even less information than that studied in Section 6.1 arises when player P_1 has knowledge

only of its own win probabilities, and not those of other pairs of players. For this case, we have the following result.

Theorem 2. *For any instance of an n -stage tournament, let players $P_2, \dots, P_{2^n}$ be ordered such that $p_{12} \leq p_{13} \leq \dots \leq p_{12^n}$. Then, if player P_1 chooses the lowest ranked available player at each round, this strategy achieves $q_1 \geq \prod_{i=1}^n p_{12^i}$, and this bound is tight.*

Proof. We prove the result by induction on the number of rounds played. Let $q_1(k)$ denote the probability that player P_1 wins all its matches at the first k rounds. For the basis where $k = 1$, player P_1 chooses to play against player P_{2^n} and wins with probability p_{12^n} . For the induction hypothesis, we assume that the result holds for rounds $1, \dots, k$, and prove that it holds for round $k + 1$. Thus, we assume that $q_1(k) \geq \prod_{i=n-k+1}^n p_{12^i}$. Observe that, at round $k + 1$, there must exist a remaining player with an index of at least 2^{n-k} . If player P_1 chooses to play against this player, then we have $q_1(k + 1) \geq \prod_{i=n-k+1}^n p_{12^i} \cdot p_{12^{n-k}} = \prod_{i=n-k}^n p_{12^i}$.

To show that the bound is tight, recall that by assumption player P_1 does not know the win probabilities of other pairs of players. Let $p_{ij} = 1$, for $2 \leq i < j \leq 2^n$. In this case, when k rounds remain to be played, the remaining players are exactly $P_1, \dots, P_{2^k}$. Therefore, there is no possibility for player P_1 to achieve a higher value of q_1 . $\square$

However, we observe that Theorem 2 does not imply that player P_1 in general maximizes q_1 by choosing to play the player against which it has the highest win probability. Remark 2 in Section 3 provides a counterexample.

7 Shirking

When players shirk, there are serious implications for the credibility and prestige of the tournament, and in some situations also for national pride. In addition, major sports are a venue for substantial betting by the public. For example, at the 2018 FIFA World Football Cup more than \$1.8 billion was bet. As a result, shirking raises the possibility of legal liability for fraud.

In this section, we study the issue of shirking under both individual and group strategy by the players. Section 7.1 reviews several famous and controversial incidents of shirking. Section 7.2 discusses the extent to which our opponent choice design can reduce shirking against individual strategic behavior. Section 7.3 contains a similar discussion for group strategic behavior.

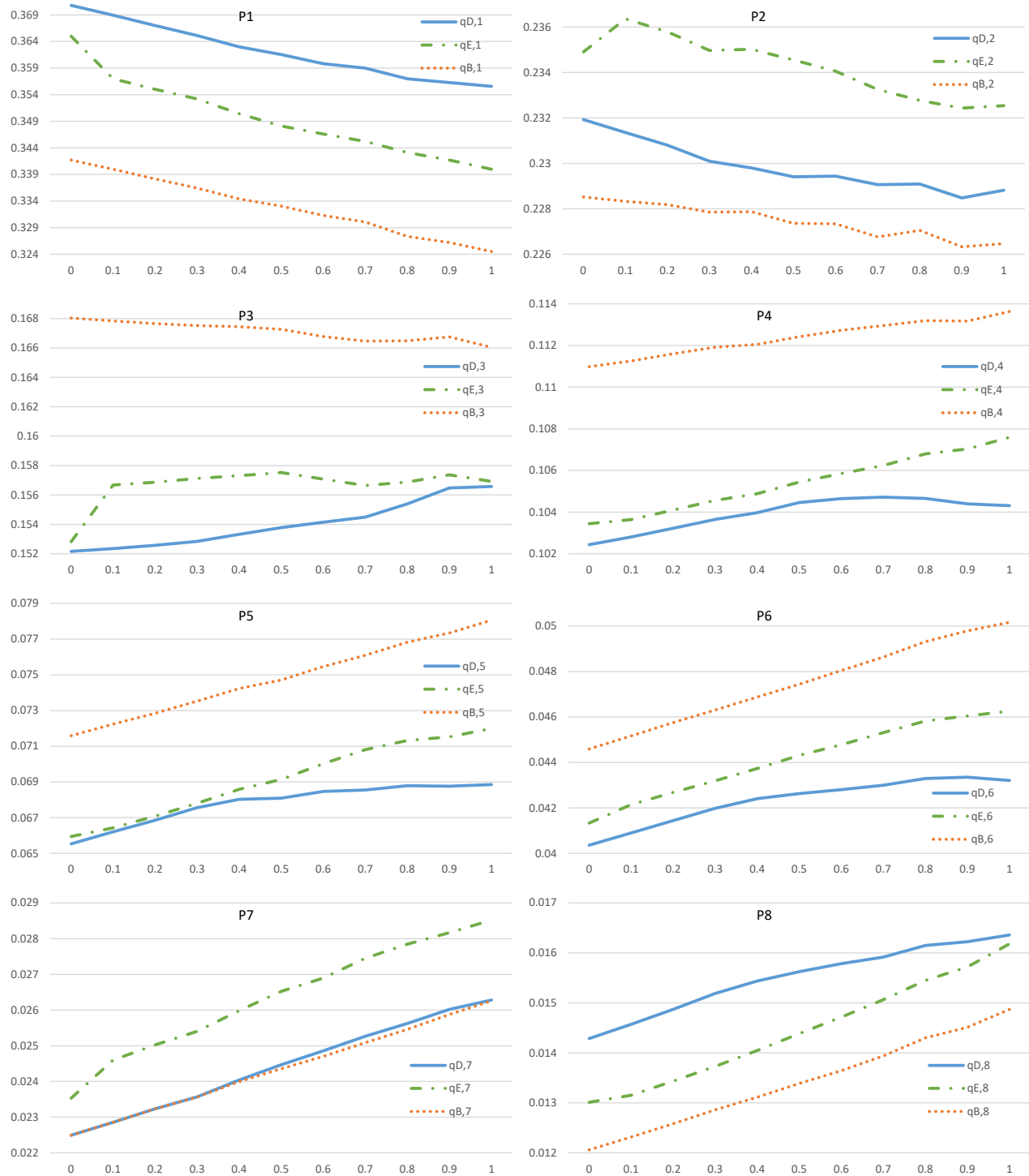


Figure 3: Tournament win probabilities with incomplete information.

7.1 Examples of shirking

We list several notorious examples of shirking in multiple round tournaments. See Vampiew (2018) and the other references shown below.

- In the 1998 Tiger Cup Asian football competition, the winner of the last group game between Thailand and Indonesia would travel to Hanoi to play the host Vietnam, whereas the loser could stay in Ho Chi Minh City and play Singapore. An Indonesian player received a lifetime ban for egregiously scoring for Thailand in extra time.
- The NBA changed its postseason tournament design after the 2005–06 Los Angeles Clippers apparently lost some late regular season games deliberately, in order to avoid facing the Dallas Mavericks in the first round of the playoffs (MetaFilter.com 2015).
- In the ice hockey competition at the 2006 Winter Olympics, the Swedish coach publicly discussed the benefit of losing an upcoming match with Slovakia in order to avoid playing either the Czech Republic or Canada, the two most recent gold medalists, at the next round. Sweden lost the match 3–0 and went on to win the gold medal.
- At the 2012 Summer Olympics, the Chinese, Indonesian and S. Korean women's badminton teams were all disqualified for deliberately losing their group stage matches, in order to achieve easier matches at the following round (Wired.com 2012).
- At the 2012 Summer Olympics, the Japanese women's football team played for a draw against South Africa, to avoid the disruption of travel for its next round match.
- At the 2018 FIFA World Football Cup, the last group match involved England and Belgium. The winner would face a potential quarterfinal match against five-time world champions Brazil. Both teams rested their top players. Belgium won the match 1–0, but later beat Brazil anyway (Deadspin.com 2018).

7.2 Individual strategic behavior

We discuss the extent to which our opponent choice tournament design reduces the players' incentive to shirk under individual player strategic behavior. We specifically consider the final game at the end of the preliminary round. While this restricts the scope of our discussion, it is in exactly this situation that shirking is most likely. Moreover, this situation includes, for example, the England – Belgium and Thailand – Indonesia games described above.

For the last group match between England and Belgium at the 2018 FIFA World Football Cup, the winner would likely

face Brazil later under the fixed bracket design. Under the opponent choice design, the winner potentially has a better opportunity to choose first, which motivates both teams to win. For the game between Thailand and Indonesia in the 1998 Tiger Cup Asian football competition, under the opponent choice design, the winner would likely receive the option to choose which game to play and avoid traveling to Hanoi to play the host Vietnam; hence shirking could be prevented.

More generally, the opponent choice design encourages winning, since winning provides a superset of choices for the next round of opponents. Additionally, the uncertainty of next round opponents under the opponent choice design reduces anticipatory shirking to avoid specific opponents. For these two reasons, the opponent choice design is typically helpful in reducing shirking.

On the other hand, in general shirking is hard to eliminate. As shown by Vong (2017), with game outcomes characterized by a network with players' choices about the amount of effort to exert, it is not possible to eliminate shirking entirely without imposing on the tournament design a severe restriction that only the top-ranked player from each group advances to the next round of the tournament. We now present two examples which show that shirking cannot be fully eliminated by the opponent choice design.

Example 1. Suppose that player P is guaranteed to continue to the next round. However, by winning a game with player O , P eliminates O but allows into the next round another player E that P cannot beat. Then, there exists a win probability matrix where P 's tournament win probability is reduced by the presence of E in the next round. As an extreme example, suppose that player E beats every player that will continue to the next round, including player P . In this case, player P will shirk in the game with player O .

In Example 1, the ranking of players other than P and O that continue to the next round is dependent on the result of the game between P and O . However, even if this condition does not hold, player P may still shirk, as shown in the following example.

Example 2. Suppose that player P is guaranteed to continue to the next round. However, by winning a game with player O , P will rank first and O will rank third. Whereas, if player P does not win, then the opposite ranking applies. Further, there exists another player E , whose ranking will be second irrespective of the result of the game between P and O . Suppose that player P has a zero chance of winning against player E , while player O can beat player E easily. Then, there

exists a win probability matrix where player O 's tournament win probability is maximized when choosing player E as an opponent, and player E can beat all players who will continue to the next round including P , except O . In this case, player P will shirk in order to allow player O to rank first and eliminate player E , else as the first-ranked player, it will not choose player E and then E as the second-ranked player will not choose player O .

Example 2 illustrates that under the opponent choice design, though typically a higher ranking provides a player a superset of choices for the next round of opponents, a lower ranking can be beneficial when it enables the use of another player as an agent to eliminate a threatening player.

7.3 Group strategic behavior

We consider possible strategic behavior by groups of players. This is most likely to occur where a tournament contains multiple teams representing the same country, for example in tennis, table tennis, or badminton at the Olympics. Note that any result that fails to eliminate shirking under individual strategic behavior also fails under group strategic behavior. Therefore, Examples 1 and 2 continue to hold under group strategic behavior.

A problematic shirking situation under a bracket design due to group strategic behavior occurs when a generally stronger player intentionally loses to a generally weaker player from the same team, in order for the weaker player to advance with a higher winning chance against an opponent with a style of play that disadvantages them. Such shirking occurs less frequently under the opponent choice design since by winning, the generally stronger player is likely to have a superset of opponents to choose from. Furthermore, the opponent that the generally weaker player will play against next is less predictable.

8 Concluding remarks

We address several deficiencies that arise in multiple round sports tournaments. These deficiencies include the existence of an unnecessary element of chance in how matchups evolve, incentives for shirking, and the inability to integrate newly developing information. To resolve these deficiencies, we recommend a new design for the single elimination stage of a tournament, where highly ranked players may choose their opponents at each round, and each player maximizes its tournament win probability by doing so. This design is implemented for both static and dynamic

rankings of the players. We describe a dynamic programming algorithm that computes, for each player, the optimal sequence of players to choose from those available, in order to maximize its tournament win probability. This algorithm is computationally tractable up to the round-of-16 or later rounds. Using a data set from 2215 men's professional tennis tournaments, we demonstrate the reasonableness of the results obtained from the opponent choice design. We also study the sensitivity of the performance of the opponent choice design to the level of irregularity in the players' pairwise win probability matrix. An additional advantage that arises from the use of the opponent choice design is increased spectator interest from the unpredictability of matchups at future rounds. Most importantly, an easier path through the tournament is typically available to any player as a result of obtaining a higher ranking, as verified by the results from our computational study in Section 5. This feature of our opponent choice design discourages strategic shirking behavior at the preliminary stage of the tournament.

We summarize the managerial insights from our work that support tournament designers. First, the inclusion of opponent choice makes a tournament particularly appealing to the highest-ranked players, who are typically the players most sought after by the tournament organizers. This is because it offers increased tournament win probability to players with high rankings, and prevents the unfortunate "bad matchup" problem. Second, the increased probability of a marquee matchup between the top two ranked players, either at the final round or earlier, is valuable to tournament designers as a significant revenue opportunity. Third, the partial elimination of shirking from opponent choice reduces the risk of negative publicity for the tournament. Fourth, the possibility that players can incorporate new information into their opponent choices adds an appealing dynamic element to the tournament. Finally, based on experience with the Austrian League tournament, it appears that substantial publicity may be achievable by a promotional event that includes players choosing their opponents.

Several interesting topics remain open for future research. We particularly recommend the following seven. The first topic applies our work, and the other six extend it.

- It would be valuable to explore the application of the opponent choice design to empirically-based studies of various sports and competitions. Examples where substantial data for pairwise matchups is available include table tennis and chess.
- Precise estimations of pairwise win probabilities present a statistical challenge, particularly when the number of participating players is large. As an

alternative approach, a player may resort to a greedy strategy: choose an available opponent that it can beat with the highest probability at each round.

- It would be valuable to study how the results of a particular round could be used to modify the win probability matrix, and consequently the choices of the players at later rounds. For example, a player that has won its matches but with unexpectedly poor performances, or has incurred an injury, may become a more attractive opponent at a later round.
- As proposed by Guyon (2022), the choice of other highly ranked players can be prohibited.
- The opponent choice design can also be used where a player's objective, rather than winning the entire tournament, is to reach a particular round. This scenario arises towards the end of a season, where a player that is seeking to be ranked first at the end of the season needs only to reach a particular round. The sequence of choices of opponents that maximizes the probability of reaching a particular round is not, in general, the same as that maximizes the player's tournament win probability (see Remark 2).
- A natural extension of the opponent choice design under dynamic ranking would be to allow the ranking of the players to be adjusted dynamically, based on detailed performance within the tournament, as measured for example by margin of victory.
- Some spectators of sports events enjoy the situation where one player is clearly favored over another, the latter often being termed an "underdog". It is intuitive that the use of our opponent choice design tends to reduce the *apparent* number of such matchups since there are fewer matches with extreme differences in rankings. However, since higher ranked players are choosing their opponents based on their actual win probabilities, the *actual* number of such matchups tends to increase. This is a tradeoff that should be evaluated by tournament designers who are in the best position to understand the interests of their spectators.

In conclusion, we hope that our work will encourage interest in this topic that influences the enjoyment of billions of sports fans around the world.

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Appendix

Example of algorithm opponent choice

We provide a numerical example of Algorithm Opponent Choice under static ranking of players for the quarterfinal round of a tournament that includes the top eight seeded tennis players in the data set described in Section 4.3. The algorithm starts from all the possible pairwise win probabilities for those eight players, which are used to define the boundary conditions at the final round. It then calculates the tournament win probabilities for each player, for all possible semifinal configurations. We assume that those calculations have already been completed, and now show how they are used at the quarterfinal round.

The algorithm calculates seed No. 1's tournament win probability for each of its seven possible choices of quarterfinal round opponent, seeds Nos. 2 through 8. For conciseness, we show in Table 11 only the detailed calculation of seed No. 1's tournament win probability if it chooses to play against seed No. 7, which is the choice that maximizes its tournament win probability. This probability is computed by the algorithm as follows. If seed No. 1 makes this choice, the other quarterfinal games, given optimal sequential selections by the next highest ranked available players are also computed recursively, which are seeds Nos. 2 and 3, respectively, are (2, 4), (3, 6) and (5, 8). Given the assumption

Table 11: Tournament win probability of seed No. 1 if seed No. 7 is its quarterfinal opponent.

i	s_i	$p(s_i)$	$p(q_1 s_i)$	$p(q_1, s_i)$
1	(1, 5, 2, 3)	0.105	0.436	0.046
2	(1, 8, 2, 3)	0.217	0.432	0.094
3	(1, 5, 2, 6)	0.066	0.443	0.029
4	(1, 8, 2, 6)	0.137	0.439	0.060
5	(1, 5, 4, 3)	0.048	0.478	0.023
6	(1, 8, 4, 3)	0.099	0.474	0.047
7	(1, 5, 4, 6)	0.030	0.498	0.015
8	(1, 8, 4, 6)	0.062	0.494	0.031
			Total	0.344

of independence, the probability of any combination of winners from those four games is the product of their pairwise win probabilities. For example, using data from Table 5, the probability that the winners of the four quarterfinal games are seeds Nos. 1, 2, 3 and 5 is $p_{17} \times p_{24} \times p_{36} \times p_{58} = 0.764 \times 0.687 \times 0.613 \times 0.327 = 0.105$. Computing the probabilities of all eight possible combinations of quarterfinal winners provides the results shown in the third column of Table 11. We note that these probabilities do not sum to 1, since only outcomes where seed No. 1 wins its quarterfinal game are considered. On the condition that the other semifinal round players are seeds Nos. 2, 3 and 5, seed No. 1 has three choices:

Play seed No. 2: $q_1 = 0.548[(0.567 \times 0.649) + (0.433 \times 0.727)] = 0.374$.

Play seed No. 3: $q_1 = 0.649[(0.632 \times 0.548) + (0.368 \times 0.727)] = 0.398$.

Play seed No. 5: $q_1 = 0.727[(0.487 \times 0.548) + (0.513 \times 0.649)] = 0.436$.

Since Algorithm Opponent Choice operates by backward recursion, these probabilities have already been computed by the algorithm at the semifinal round, and are known to seed No. 1 at the time of choosing its quarterfinal opponent. In view of seed No. 1's objective of maximizing its tournament win probability, it chooses to play seed No. 5 at the semifinal round. There are eight possible semifinal configurations in which seed No. 1 is included, as shown in the second column of Table 11. We denote these eight possible semifinals by s_i , $i = 1, \dots, 8$. The probabilities of their occurrence appear in the third column as $p(s_i)$. The conditional probability for seed No. 1 to win the tournament from each semifinal appears in the fourth column as $p(q_1|s_i)$. The fifth column of Table 11 shows the joint probability $p(q_1, s_i) = p(s_i) \times p(q_1|s_i)$ of each possible semifinal configuration i in the row and seed No. 1 winning the tournament from that configuration. Finally, the probability of seed No. 1 winning the tournament, still conditional on choosing seed No. 7 as its quarterfinal opponent, appears at the bottom of the fifth column of Table 11, where $q_1 = \sum_{i=1}^8 p(q_1, s_i)$.

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